Curve25519: matched isogeny, carry and rank instruments

2026-11-04. Development implementation; experiment comparisons are pending. All observations in this report are producer assertions [pcc:unverified].

What is implemented

1. **Isogeny support controls.** The native instrument constructs source, transported-identical, target-native, mixed and random factor bases with equal numbers of sign-orbit columns. A GMP fixture exercises the actual RFC Curve25519 generator, subgroup, rational 2-isogeny, dual and inverse transport. The finite atlas experiments themselves are small-prime diagnostics; no 255-bit relation experiment is implemented or reported.
2. **Matched relation encodings.** Two QF_NIA emitters describe the same signed, distinct-column, three-summand group query: direct modular multiplication and radix-bounded schoolbook multiplication/reduction carries. Full affine addition includes infinity, doubling and inverse branches. Membership is an explicit point table. This is not compact membership or a new summation-polynomial construction.
3. **Training-only rank selection.** A generic decomposition enumerator replays hits by the group law and computes coefficient rank over ell. Candidate selection uses training rank per pair probe; disjoint holdout queries are reported separately. It does not optimize or measure full pipeline cost yet. The emitted rank is not target-recovery success.

Editable diagram: workflow.dot; vector: workflow.svg. Dashed routes are pending. This is a workflow diagram, not a computed isogeny graph, and contains no measured speed values.

The RFC model identity already in `docs/curves/registry.json` is `icv1-fp255-tm2219385422189788282868155023270691879 62-5eb6d947`. No destination registry identity is asserted for the development quotient.

Source review and novelty correction

Read before choosing this implementation:

- RFC 7748 section 4.1: exact Curve25519 parameters and Montgomery/Edwards equivalence.
- Galbraith, chapter 15, sections 15.8–15.9: summation-polynomial factor bases, difficult prime-field decomposition and extension-field descent assumptions. The source was read in the preceding review; this is not an exhaustive current-literature search.
- crypto-autoresearcher lead atlas: `LEAD-ICP-isogeny-transported-relation-yield`, `LEAD-STR-class-representative-ic-selection`, `LEAD-ICP-digit-presentation-solving-degree`. Respectively these already track transported supports, held-out representative selection and carry presentations. See `knowledge/leads/ecdlp/{ic-prime,structure}/` there.
- Curve traits source and walk implementation: splitting traits are distinct from construction of an executable map. The independent development fixture does not extend the registry or ecdb.
- Round-7 Edwards systems: existing E1/E1Q/TW/TQ encodings overlap the proposed representative search; do not count this study as a new version of that result.
- Existing tower implementation gap: a finite point-table query does not implement the missing scalable isogeny-tower algebraic membership oracle.

The original chat proposals were not audited for novelty. Repository retrieval found direct precedents for all three. The deliverable here is a matched development instrument and a prospective combination, not a claim that the mechanisms have not been tried [pcc:unverified].

Development checks and retained failures

Build with GCC/C++20 and GMP:

```
make test
bash test-products.sh /tmp/curve25519-atlas-carry-rank /path/to/z3 /tmp/fresh-products
bash test-encodings.sh /tmp/curve25519-atlas-carry-rank /path/to/z3 /tmp/fresh-relations
```

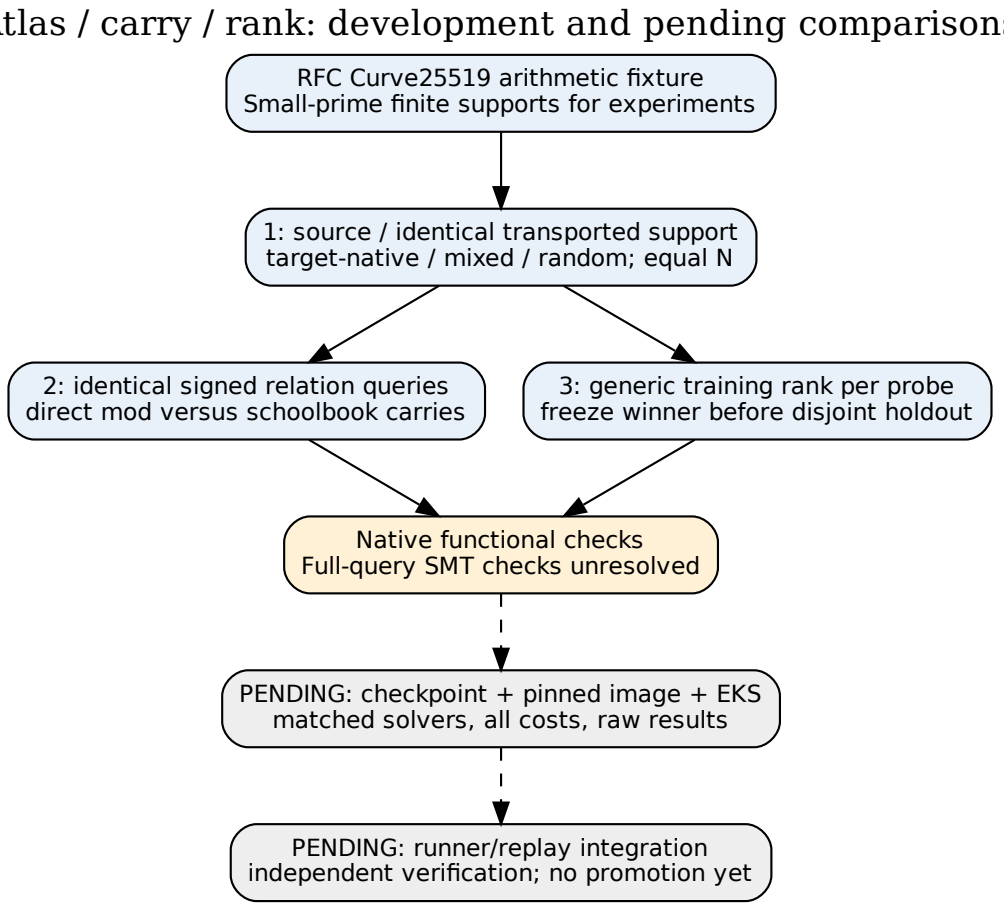


Figure 1: Prospective comparison and evidence boundaries

`make test` covers the RFC fixture, exhaustive small-curve homomorphism/dual checks, inverse subgroup transport, carry witnesses, dependent/independent rank rows, identical-support collection, and malformed-model rejection. See development checks for exact commands, raw output, versions, scope and failures. Passing a producer's functional checks is not independent verification [pcc:unverified].

External relation-encoding tests remain unresolved: initial direct-modular UNSAT fixtures hit their 30-second development test limit. Both the p=1019 initial case and the p=101 follow-up are retained, as is the attempt after removing redundant curve equations. No timeout is called UNSAT, and no unexecuted arm is called a result. Product-only tests isolate the carry component; they do not replace the full relation checks.

Execution and remaining obligations

The requested work is **partial**: all three instruments exist, with the checks scoped above; matched EKS comparisons and independent replay have not run. No performance or scaling result is claimed.

PROTOCOL.md freezes the proposed panel, controls, classification, budgets and stop rules. Its execution state is review_required. The checkpoint upload and authenticated custom-image run route are unresolved; see the additive launch-gate record. The ordinary GitHub workflow builds ecbench, not this instrument. No paid experiment was launched, and no job continues in the background.

Next technical work: integrate small-prime SAT-model replay into the frozen solver runner, satisfy checkpoint/image/route admission, run the unchanged panel, then obtain independent verification. Keep the actual 255-bit solver integration and full-cost rank-selection comparison open.

No new verified graph or numeric leaderboard finding is asserted. The canonical curve/isogeny registry and `docs/index-calculus-scoreboard.html` are left unchanged; this study adds a separately labeled prospective diagram. No Lean theorem is claimed: this is executable finite development checking, not a new universal identity proof [pcc:unverified].